

Joyvishek Ghosh

Module Lead — AI Architect

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PROFILE SUMMARY

AI Architect and Module Lead with 8+ years of product engineering experience. Architects the enterprise AI layer at ARC Document Solutions — agentic orchestration, GraphRAG, MCP tool servers, long-term memory, voice agents and document / computer-vision AI — while leading a 12-15 member engineering team. Focused on turning AI prototypes into secure, grounded, production-grade systems that work over real enterprise data.

TECHNICAL SKILLS

Generative AI	LLM Orchestration, Agentic AI, RAG, GraphRAG, MCP, Tool Calling, Prompt Engineering, AI Guardrails, OpenAI, Gemini, Ollama, Hugging Face, Rasa, LiveKit
Vision / Doc AI	YOLOv8/v11, Object Detection, Instance Segmentation, VLMs, OCR (Surya, Tesseract), OpenCV
Data / Retrieval	Neo4j (Cypher), Qdrant, pgvector, PostgreSQL, MongoDB, Redis, ClickHouse, Elasticsearch, Solr
Backend	Python, FastAPI, Node.js, NestJS, Kafka, gRPC, Socket.IO, Docker, Microservices, Event-driven Architecture
Frontend	React.js, Next.js, Angular, TypeScript, Micro-frontends, PSPDFKit, PDF.js

WORK EXPERIENCE

Module Lead — AI Architect

Sep 2024 – Present

ARC Document Solutions, Kolkata

Enterprise AI Assistant & Orchestration

- Architected **ARC Facilities AI**, an enterprise assistant that answers natural-language questions over buildings, assets, drawings, documents, forms and locations — every answer or action backed by graph data, vector search, internal APIs or memory.
- Built the orchestration layer that routes each request between Q&A, graph retrieval, document lookup, location search, data export and action tools, forwarding authenticated user context to every call.
- Migrated a monolithic RAG application into **9 independent MCP tool servers** — Graph, Actions, Document, Memory, Maps, Search, Export, Analytics and Guardrail — without breaking existing product flows.
- Standardised the servers on typed tool schemas, a shared tool registry, structured errors, health checks and runtime discovery, so one failing service never takes down the conversation.

GraphRAG, Memory & Voice

- Designed **GraphRAG** retrieval over Neo4j/Cypher and Qdrant: hybrid search finds relevant entities, graph traversal expands connected context, and the LLM (OpenAI, Gemini, Ollama) answers only from that evidence.
- Used Redis as a distributed cache for session, history and retrieval state, and Kafka for event-driven messaging between AI agents and core services — lowering latency and LLM cost.
- Added long-term user memory with confirm / update / delete controls, conversation history with paging, and session continuity, cutting token usage on repeat questions.
- Built a real-time **AI voice agent** on LiveKit with speech-to-text, turn detection and noise cancellation, connected to facility search and the main assistant's history.
- Built custom NLP components — feedback sentiment analysis and a context-aware spell checker for form data.

Computer Vision & Document AI

- Built OCR pipelines (Surya OCR, Tesseract, PDF text layers) for blueprints, technical manuals and facilities documents.

- Trained and deployed YOLO models for building annotation and equipment detection, and YOLOv11 instance segmentation for floor-area and structural-boundary isolation.
- Designed **architectural-drawing intelligence** that reads drawing sets (title blocks, legends, keynotes, schedules) using a VLM, knowledge graph and hybrid search, and cites the exact sheet in each answer.

Security & Production Readiness

- Led hardening before production rollout: closed authentication gaps, enforced user isolation, added prompt-injection guardrails, removed hard-coded secrets, set input-size limits and fixed dependency vulnerabilities.
- Added retries, fallbacks and structured logging so the assistant degrades gracefully when memory, analytics or external services are unavailable.

Product Engineering & Leadership

- Led a 12–15 member engineering team across requirements gathering, architecture, delivery and production support.
- Implemented role-based access control across Equipment, Compliance, Forms and Pin; led the Compliance 1.0 → 2.0 data migration and added dynamic surveys.
- Upgraded PSPDFKit in Forms (share links, report downloads, 1.0 → 2.0 conversion) and built a PDF viewer/signing tool with pdf-lib, PDF.js and Signature Pad.
- Built a Mutation Observer bridge for real-time React (host) ↔ Angular (remote) micro-frontend communication; integrated Google Maps for campus location search.
- Built NestJS services (clean/onion architecture) with PostgreSQL and Redis — Equipment bulk upload, Forms activity tracking, recurring task scheduling and email notifications.

Full Stack Developer

Jun 2023 – Sep 2024

Trigent Software (client: ARC Document Solutions)

- Built client and server applications with React, Next.js, Redux, NestJS, Node.js, PostgreSQL and MongoDB, including YOLOv8 computer-vision features.
- Shipped features that drove a **20% increase in user engagement**; resolved 15% more technical issues within deadlines and improved project completion time by **30%** through agile practices.
- Integrated Redis, Socket.IO, Solr and Elasticsearch; used Molecular microservices for job/worker nodes and Turbo.js for micro-frontends.

Artificial Intelligence Consultant

Feb 2021 – Mar 2023

Pemari Technology Pvt. Ltd.

- Built a Rasa-powered conversational AI chatbot for Broadcom's Clarity PPM, with a React front end and a Meteor/Node.js admin panel.
- Designed rich chatbot responses — tables, charts, images, video, guided flows, ratings, buttons and carousels.
- Connected the admin panel to Rasa through GraphQL/Apollo Server; added multitenancy for per-client data isolation and multiprocessing for performance.
- Set up automated Docker image builds into a private registry; led delivery on a 6-person team.

Full Stack Developer

Mar 2019 – May 2019

Fleapo Corporation

- Built web applications including Realbooks and a T-shirt e-commerce site with Angular, Node.js and MongoDB.
- Worked directly with customers on progress and feedback; reviewed code, debugged issues and shipped fixes.

Full Stack Software Developer

Mar 2017 – Mar 2019

TheoreX Edutech LLP

- Developed and maintained applications in AngularJS, Node.js, MongoDB, HTML5 and CSS3; reviewed code and debugged issues.
- Worked with clients across project stages and with engineers to improve software/hardware interfaces.

PERSONAL R&D PROJECTS

- **arch3D — Drawing to 3D/BIM Model.** Converts architectural drawings (PDF, image, DXF/DWG) into editable 3D/BIM models (GLB, IFC4) through a 12-stage pipeline — OCR, wall/door/window detection, scale resolution and room reconstruction — with confidence scores and human review. Geometry comes from measured data, never from the LLM. *Next.js, NestJS, FastAPI, OpenCV, Tesseract, IfcOpenShell, PostgreSQL, Redis.*
- **AutomationOS — AI Testing Platform.** QA automation platform that runs Excel-based test suites against an AI assistant, keeps multi-turn follow-up sessions, streams live results over SSE, and auto-validates responses with an LLM judge (Ollama) using configurable QA rules and similarity scores. Includes voice test automation via LiveKit and pass/fail and latency reports. *FastAPI, React, PostgreSQL, Redis, Celery, Docker.*
- **Cognitive Runtime (USCR).** Domain-agnostic runtime that turns plain-language questions into Neo4j, SQL or vector queries, driven by configurable ontology packs, with multi-tier memory and planning/reasoning agents. *FastAPI, Neo4j, Qdrant, Redis.*
- **AgentSync.** Protocol for AI agents to negotiate and complete real-world actions with secure sessions, policy checks, confirmation gates and audit logs. *FastAPI, React, Qdrant, Redis, Celery.*

EDUCATION

PG Diploma — International Institute of Information Technology, Bangalore	2023
B.Tech / B.E. — West Bengal University of Technology (WBUT)	2017
ISC (12th) / ICSE (10th) — CISCE	2013 / 2011

LANGUAGES

English, Hindi, Bengali